

Executive classifier decision guide

State length × question count × quality × cost

Fixed labels

ModernBERT-large + heads

economics depend on utilization

Runtime labels

GLiClass Modern Large v3

self-hosted flexibility

Managed API

TypeSafe.ai JEV Noul

usage-priced; no idle GPU

Mark Austin

Independent synthetic evaluation · September 20, 2026

Executive decision map: economics versus runtime flexibility

Reference workload: 6,000 state tokens × 25 questions · self-hosted cost at 100% H100 utilization

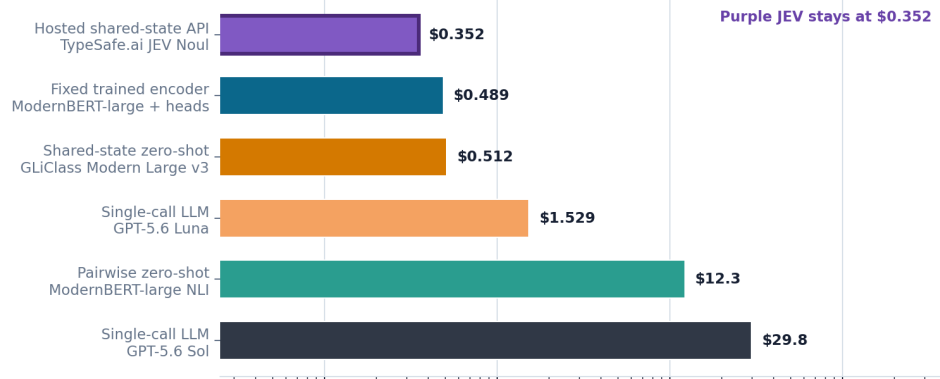


The \$1/1,000 boundary is an executive guide, not a universal purchasing threshold. Flexible systems accept new questions without retraining; trained heads do not.

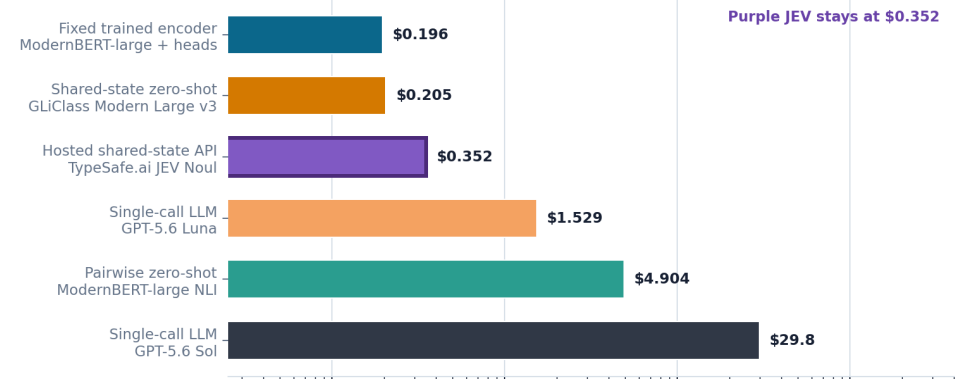
GPU utilization determines self-hosted encoder economics

Fixed workload: 6k state tokens × 25 questions · NVIDIA H100 at USD 5/hour · lower is better

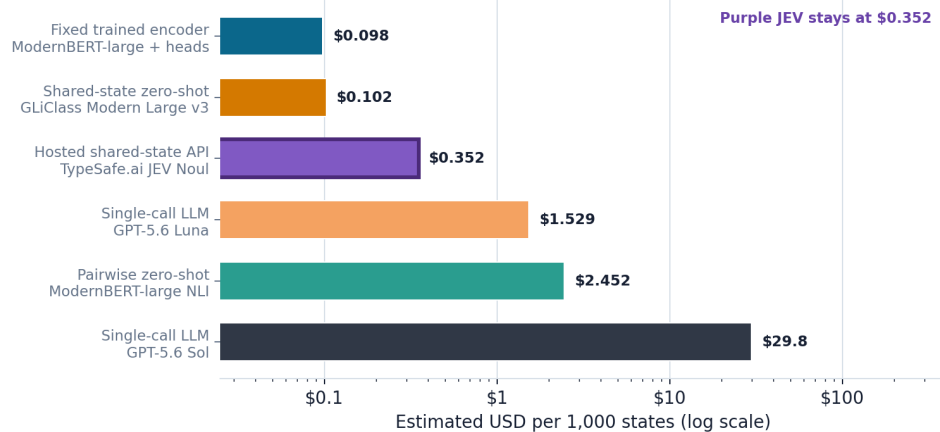
10% paid H100 utilization



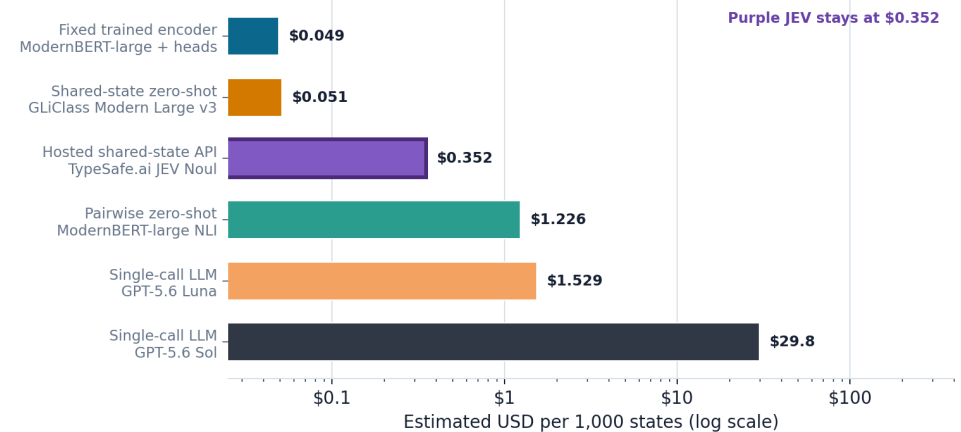
25% paid H100 utilization



50% paid H100 utilization



100% paid H100 utilization



Self-hosted encoder cost is divided by paid GPU utilization; hosted JEV/Luna/Sol prices remain usage-based. In this scenario, JEV crosses trained ModernBERT near 13.9% utilization and GLiClass Modern near 14.6%. Hidden LLM reasoning tokens are excluded.

What should an executive choose?

Select the operating model before selecting the model

Known, stable taxonomy

ModernBERT-large + trained heads

Best modeled economics with 99.16% test accuracy when utilization is sustained. New questions require labeled data and retraining.

Runtime labels, self-hosted

GLiClass Modern Large v3

Best open zero-shot trade-off tested. In the reference workload it becomes cheaper than JEV above about 14.6% paid GPU utilization.

Runtime questions, managed service

TypeSafe.ai JEV Noul

99.90% calibrated accuracy and no customer-owned idle GPU cost. It leads the reference scenario below roughly 14% utilization.

Broad reasoning or maximum test score

Luna or Sol

Near-ceiling quality, but use when semantic breadth justifies the premium. Sol's advantage over calibrated JEV was one decision in 4,050.

Executive guardrails

- Quality results come from a synthetic, explicit-evidence test and are not production guarantees.
- Cost curves are modeled from token shapes, published/API rates, and H100 throughput proxies.
- Validate with real states, real questions, target hardware, and operational error costs before procurement.